AdaTTT: Adaptive Test-Time Training for Efficient Visual Question Answering

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1. Dataset. Our primary dataset is **VQA-v2**, the standard VQA benchmark, providing over 1.1 million question–answer pairs across 200,000+ MS-COCO images with three question types: yes/no, number, and open-ended. We also evaluate on **VizWiz**, containing 31,000 questions from photos taken by blind users, which exhibits natural distribution shift through low-quality images and an "unanswerable" class—making test-time adaptation particularly impactful. As a cross-task evaluation, we apply our framework to **Memotion2**, a multimodal meme dataset with OCR-extracted text and sentiment/humor/sarcasm labels. Memes exhibit natural distributional shift as formats evolve, ideal for testing whether adaptive TTT generalizes beyond VQA.

2. Task/Problem. Given input $\mathbf{x} = (I, Q)$, the system predicts answer $\mathbf{y} = A$ via classification over the top 3,129 most frequent answers. For Memotion2, $\mathbf{x} = (I, T)$ where T is OCR text, and $\mathbf{y}$ is a sentiment label. Images are resized to 224×224 with ImageNet normalization; questions use BERT WordPiece tokenization. Feature extraction uses frozen ViT-B/16 for visual embeddings and frozen BERT-base for text encoding. The pipeline runs on Colab Pro (T4/A100). The core problem is **when and how to adapt at test time**: standard TTT applies expensive gradient updates to every sample, but many "easy" samples gain no benefit. We introduce a learned confidence gate that routes this decision per-sample, enabling a principled accuracy-vs-compute tradeoff.

3. A Plausible Deep Learning Pipeline. The pipeline has four stages with three sets of learnable parameters. **(1) Encode:** Frozen ViT-B/16 produces 197 visual patch tokens; frozen BERT-base produces text tokens. Both encoders are fixed with zero trainable parameters. **(2) Fuse:** A 2-layer bidirectional cross-modal attention module (θ_f, $\sim$4.7M params) performs cross-attention in both directions with residual connections and layer normalization, producing a pooled representation $\mathbf{z} \in \mathbb{R}^{768}$. **(3) Gate:** A lightweight MLP (θ_g, $\sim$50K params) maps $\mathbf{z}$ to a confidence score in $[0, 1]$. Samples exceeding threshold τ take the fast path (skip TTT); the rest route to adaptation (slow path). **(4) Predict:** For skipped samples, an MLP head (θ_d, $\sim$2.3M params) predicts from $\mathbf{z}$. For adapted samples, K self-supervised gradient steps update θ_f and θ_d using masked patch prediction—masking 25% of visual tokens and reconstructing them—producing adapted $\mathbf{z}'$ for prediction. Parameters are restored after each sample to prevent cross-sample drift. The architecture is task-agnostic: only θ_d changes between VQA and sentiment. The central deliverable is the **Pareto frontier of accuracy vs. GFLOPs** across $K \in \{0, 1, 2, 3, 5\}$ and $\tau \in \{0.5, 0.7, 0.8, 0.9, 0.95\}$.

4. Plausible Regularization. TTT risks catastrophic forgetting when gradient steps push parameters too far from their trained values. We address this with two complementary strategies applied **within the TTT loop**. **Consistency regularization:** two augmented views of the test image (random crop + color jitter) are generated per TTT step, and a symmetric KL penalty ensures stable predictions across views, discouraging the model from becoming sensitive to superficial input variations. **Mixup anchoring:** the adapting representation is interpolated with the pre-adaptation anchor $\mathbf{z}_0$ as $\mathbf{z}_{\text{mix}} = \alpha\,\mathbf{z} + (1-\alpha)\,\mathbf{z}_0$ with $\alpha \sim \mathcal{U}(0.7, 1.0)$, constraining drift to a neighborhood of the original representation. Both techniques serve TTT stabilization; they are not independent strategies. We conduct a structured ablation: TTT alone vs. TTT+consistency vs. TTT+mixup vs. TTT+both. Motivated by Karmanov et al. and Farina et al., who showed constrained adaptation improves VLM robustness under distribution shift.

5. Group Unification. The pipeline is modular and task-agnostic, enabling clean parallel development. **Aishwarya** develops the base VQA system: frozen encoder integration, cross-modal fusion module design, supervised training of θ_f and θ_d, and TTT stabilization techniques. **Yugesh** develops the TTT adaptation loop (masked patch and rotation objectives), confidence gate training using oracle labels that compare base vs. TTT predictions, and evaluation infrastructure including Pareto analysis. **Aryan** evaluates AdaTTT on Memotion2 meme sentiment classification, testing cross-task generalization by swapping only the prediction head while reusing the shared frozen encoders and fusion module.

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